PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

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```
from collections import deque
def water_jug(cap_x, cap_y, target):
    start = (0, 0)
    queue = deque([(start, [start])])
    visited = {start}
    while queue:
        (x, y), path = queue.popleft()
        if x == target or y == target:
            return path
        pour_xy = min(x, cap_y - y)
        pour_yx = min(y, cap_x - x)
        moves = [
            (cap_x, y),
            (x, cap_y),
            (0, y),
            (x, 0),
            (x - pour_xy, y + pour_xy),
            (x + pour_yx, y - pour_yx),
        ]
        for s in moves:
            if s not in visited:
                visited.add(s)
                queue.append((s, path + [s]))
    return None
cap_x, cap_y, target = 4, 3, 2
print(f"Jugs: {cap_x}L and {cap_y}L, target: {target}L")
for step in water_jug(cap_x, cap_y, target):
    print(step)
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Jugs: 4L and 3L, target: 2L
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
>>>
```